

## Category: ENGINEERING

**SOFTWARE ENGINEERING MANAGER Summary** Experienced software engineer, and hands-on engineering manager with 20+ years of product development experience in broadcast media, and 10+ years of experience in leading development teams (20+ engineers), hiring (10+ offers, 40+ interviews), HR personnel procedures, performance reviews, and project transitions. Experience 09/2009 to Current Software Engineering Manager Company Name ■ City , State Visual C++, Windows, STL, OOP, MFC, threads, file maps, ATL, IPC, FTP, TCP, HTTP, XML, JSON, services, web services, REST API, SOA, media formats and standards, codec SDK and integration, MPEG-DASH, API design and documentation. BCD spinoff as Imagine in 2013) Nexio video servers and applications facilitate collaborative workflow in Nexio LAN with video servers, services and applications running in various nodes. Media comes from proprietary SAN or generic storage off NAS. Development and team management, Nexio servers' applications, Playout business unit: Relocated from IL to CA, took on larger role, and video server applications as well. Led team of 6 from Burbank, team of 8 from Naperville, and team of 8 (indirect report) from Beijing. Introduced peer reviews, repurposed components, and all 3 teams collaborated as one large team. Adopted Scrum since 2014 (ScrumMaster), team of 3 to 6 from Burbank; Naperville office closed in 2013. Team migrated source revision control to Mercurial / Jenkins, 30+ products. Recipient of "Reuse Innovation Award" for IP reuse, major factor to Server business unit turnaround. Released deliverables for Nexio Software Suite 6.0, 7.0, 8.0, Global Proxy Suite 2.5, 3.0, 3.5 Initiated, architected and executed SOA from concept to completion to facilitate decoupled UI clients, 2 web services and 10 services, and selected hands-on projects include: Hercules web service for web player solution based off MPEG-DASH Created C++ engine to process URI, handle sessions, and deliver segmented MP4 atoms (H.264, AAC) Provides dynamic HTML5 player page; JavaScript source based off dash JS. Supports multiple raster sizes and bit rates; "ready to go" and "just in time" sessions. Team helped with providing C# web service framework, CLI bridge layer, and device testing. Published functional, REST API, and URI specification document. JSON services on TCP stack using C++ as part of SOA Created API framework classes; multi-threaded server side socket classes. Created header only JSON parser on top of open source "rapid json" for SAX style reader. Developed new services: Content Manager, Helper, Picon, and Requestor. Evolved as services: Encoder, GPRX, Helios, MB, Scavenger, and Transcoder. Guided GPRX, Helper, and Requestor to completion. Published 10 JSON protocol API documents for 3rd party and in-house use. Content Manager service for primary-backup workflows Multi-threaded rules based engine to manage media contents between two domains. Architected and developed new UMID based solution (gen 2) for new workflow needs and scalability. Scavenger, Transcoder for low-res proxy management Needed solution to improve legacy products, required regular hand holding, remained escalated. Proposed new approach, teamed up, and successfully wrote two new 24/7 products. MediaBase DLL, desktop explorer for assets (C++, MFC) Guided to optimize original implementation for 200K IDs to meet storage scalability needs. Achieved constant UI performance under few milliseconds regardless of user operation. Proposed to introduce "dta-handler" architecture in ftp-server Approach helped to consolidate all media formats as one product build, and scalability. Team added one converter DLL per media container format for import and export. 11/2005 to 08/2009 Software Engineering Manager Company Name ■ City , State Development and team management, Nexio editing applications: Led team of 8 from Naperville, and team of 8 (indirect report) from Beijing. Hired, mentored and built core editing team to size of 8, and collaborated with expanded team in Beijing. Established development process document for consistent procedures and peer code review guidelines. Team established crash dump analysis procedures to help troubleshoot hard to repeat issues. Released deliverables for Video Editing Suite 2.5 Proposed architecture and teamed up to develop server integration modules for video broadcast workflows. 08/1997 to 10/2005 Principal Software Engineer Company Name ■ City , State Harris Corporation acquired Leitch) Architected, designed, and implemented fundamental DLL components for evolution of video editing product, and major projects include (C++, Win32, threads, GUI): Interactive playback architecture Media file reader and writer components, scalable decoder and encoder architectures. Video and audio rendering engines Hardware integration modules, critical part of business success in post-production space. Interactive picon and waveform drawing components 64-bit and Unicode migration of all modules with over 4 million lines of code. 08/1994 to 07/1997 Software Engineer Company Name ■ City , State DPS acquired Star Media;

Leitch acquired DPS in 2000). Star Media, technology startup, created video editing software Velocity for post-production space, and developed projects include Project media management tools, Render Bank, and video effects. Education M.S : Computer Science Illinois Institute of Technology ■ City , State Computer Science B.E : Electronics and Communication Engineering National Institute of Technology ■ City India Electronics and Communication Engineering Products: <http://www.imaginecommunications.com/products/playout/video-servers> Skills API, Approach, ATL, audio, backup, broadcast, C++, CLI, com, CA, Hardware, concept, Content, clients, documentation, dynamic HTML5, editing, XML, FTP, functional, drawing, GUI, http, IDs, Innovation, explorer, IP, JavaScript, json, LAN, MB, C#, MFC, office, Win, Windows, 2000, migration, NAS, OOP, page, Proxy, rendering, SAN, Scrum, servers, specification, team management, threads, troubleshoot, Video, Video Editing, Visual C++, workflow, writer